



Carbon Monoxide

Assessment

Pediatric Pearls:

- Use pediatric dosing of medications or electrical therapy for a pediatric patient < 37 kg and as defined by the PEDIA Tape.

Signs & Symptoms:

- Altered mental status/dizziness
- Headache, Nausea/Vomiting
- Chest pain/Respiratory distress
- Neurological impairments
- Vision problems/Reddened eyes
- Tachycardia/Tachypnea
- Arrhythmias, seizures, coma

Differential:

- Effects of other toxic fire byproduct (ie. Cyanide)
- Acute cardiac event
- Acute neurological event
- Flu/GI illness
- Acute intoxication
- Diabetic Ketoacidosis
- Headache of non-toxic origin

Clinical Management Options

P	P	P	P	P	P	• Measure COHb% (SpCO), if equipment is available with covered or shielded probe
L	L	L	L	L	L	• <u>SpCO > 25% with or without red flag symptoms:</u> Utilize COGs as indicated, provide airway/ventilation management with 100% supplemental oxygen, and transport to an appropriate facility. ○ Red flag symptoms include: Pregnancy*; any cardiac or pulmonary complaint; Neurological changes such as altered mental status, seizure activity, or focal deficits. A headache alone is not considered a neurological change or red flag symptom. • <u>SpCO 16 – 24% with symptoms but no red flag symptoms:</u> Remove the patient from the source and treat with 100% supplemental oxygen until symptoms improve or SpCO decreases to < 15% then transport or patient refusal as appropriate. • <u>SpCO < 15% and asymptomatic:</u> No treatment is required. ○ Any patient presenting with a red flag symptom, regardless of SpCO, is to receive airway/ventilation management with 100% supplemental oxygen, and transport to an appropriate facility.
1	2	3	4	5	6	• Continuous ETCO₂ and ECG monitoring • Acquire 12 lead ECG
						• Vascular access as appropriate for patient condition • IV fluid with isotonic crystalloid as needed
						• Fire victims with AMS, lethargy, or cardiac arrest, consider Cyanide Guideline

Consult Online Medical Control As Needed

Pearls:

- Refer to drug formulary charts for all medication dosing for both adults and pediatric patients.
- *Fetal hemoglobin has a stronger affinity to CO than maternal hemoglobin. Therefore, transport to an ED of a known/suspected pregnant female patient should occur for this reason.*
- The absence (or low detected levels of) of COHgb is not a reliable predictor of firefighter or victim exposure to other toxic byproducts of fire.
- The differential list for CO Toxicity is extensive. Attempt to evaluate other correctable causes when possible.
- Chronic CO exposure is clinically significant; therefore advice on smoking cessation is important medical instruction and; recommend evaluation of their home/work environment for presence of CO.